SANJAY PADMARAJ

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EDUCATION

New Horizon College of Engineering, Bangalore, India — 2025 – 2029 (expected)
Bachelor of Engineering (B.E.), Computer Science & Engineering — CGPA: 9.75 / 10

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C, C++, SQL, HTML/CSS
Web & Backend: React, Next.js, Node.js, Express, FastAPI, Flask, Tailwind CSS, PostgreSQL, Supabase, Socket.IO
AI / ML: RAG, LLM and multi-agent apps (LangGraph), Hugging Face, Sentence-Transformers, scikit-learn, Pandas, NumPy
Tools & Testing: Git, GitHub Actions (CI), Docker, Jest, Vitest, Playwright

PROJECTS

CampusOS — unified campus-life platform — github.com/123sanjaypadmaraj/CampusOS
React 18, JavaScript/TypeScript, Vite, Tailwind CSS, TanStack Query, Supabase (PostgreSQL, Auth, Edge Functions)

- Built a full-stack platform that consolidates six campus workflows (events, clubs, community feed, hackathon team-matching, resource booking, issue reporting) otherwise spread across chat groups, forms and separate portals.
- Designed the PostgreSQL schema and Supabase backend; built Razorpay order and webhook handling in Edge Functions (test mode).
- Set up a five-stage GitHub Actions pipeline (lint, typecheck, unit tests, build, end-to-end) with Jest and Playwright.

Kotlex — multi-agent marketing campaign builder (HACKERING 2.0) — github.com/hackering-2-0/HR2-AI-F0C5415B
Python, FastAPI, LangGraph, Server-Sent Events, Pillow, Tesseract OCR, pytest

- Built a round table of 12 LLM agents that turns a café owner's voice note (Kannada, Hindi or English) into multilingual copy, a poster and a reel, orchestrated with LangGraph and streamed live to the browser.
- Wrote a code-only Fact Guard that blocks drafts whose discount, price, days or times differ from the owner-confirmed facts, a dependency graph that regenerates only the assets a change affects, and an LLM router with fallback; 500+ tests in CI.

Club Hub — multi-club management app — github.com/123sanjaypadmaraj/Club-App
Next.js, TypeScript, Supabase (PostgreSQL, Auth, Row-Level Security), Vitest

- Built a three-role app (visitor, club lead, admin) for a college's co-curricular clubs: event registration with waitlists and QR tickets, check-in, feedback, CSV exports, and an admin dashboard comparing club performance over 3/6/12 months.
- Restricted participant data (names, emails, phones) with Postgres row-level security so only a club's leads and the admin can read it, even through direct API calls; made event-capacity checks race-safe in SQL.

AI WordPress Builder — chat-driven website builder — github.com/123sanjaypadmaraj/AI-Wordpress-Website-
TypeScript, Docker Compose, WP-CLI, Playwright, LLMs

- Built an agent that turns a chat brief into a structured site specification, then builds a real WordPress site (theme, pages, navigation, plugins) through WP-CLI inside an isolated per-project Docker Compose stack.
- Routed every tool call through one dispatcher enforcing permission tiers, theme/plugin allowlists and audit logging; added checkpointed chat edits with undo (database restore) and automated Playwright smoke tests.

C Visualizer — interactive C learning tool — github.com/123sanjaypadmaraj/Code-Visualizer
Next.js, TypeScript, Tailwind CSS, CodeMirror, Vitest, Playwright

- Wrote a C interpreter in TypeScript that runs in the browser and draws arrays, pointers, structs, stack and heap step by step, with a trace table and a plain-English explanation of every step.
- Added live syntax-error detection with one-click fixes and optional AI autocomplete that falls back across four LLM providers; tested with Vitest (including a gcc output comparison) and Playwright.

Aero Engine Digital Twin — Smart India Hackathon prototype — github.com/123sanjaypadmaraj/Aero-piston-engine-simulation
Node.js, Express, Socket.IO, three.js, JavaScript

- Simulated telemetry for a 3-UAV fleet (13 sensors per engine, 9 injectable fault scenarios), streamed over Socket.IO every 2 seconds to a live dashboard and a three.js 3D engine model.
- Implemented fault prediction from rolling z-scores and threshold rules, producing a 0–100 health score, an estimate of remaining useful life and a predicted fault type without an external ML runtime.

Smart Bandage — wearable biosensor software stack — github.com/123sanjaypadmaraj/Smart-Bandage
Python, FastAPI, PostgreSQL, WebSockets, React, React Native, Docker

- Built the signal-processing pipeline, FastAPI backend, React dashboard and React Native app against a simulated sensor with 7 fault scenarios; added anomaly detection, alerts and Gemini-backed insights (159 tests in CI).

Accident Detection Hardware — ESP32 sensor node — github.com/123sanjaypadmaraj/AI-Traffic-Detection-Hardware
C++, ESP32, PlatformIO, MPU6050, GPS, ultrasonic sensor

- Wrote ESP32 firmware reading an MPU6050 IMU, GPS and ultrasonic sensor that flags a possible accident from total acceleration.

EXPERIENCE & LEADERSHIP

STEM Club, New Horizon College of Engineering — Member, website and AI chatbot — 2025 – 2026

- Built and maintain the club's official WordPress site on the college domain (newhorizonindia.edu/co-curricular/stem).
- Built a retrieval-augmented (RAG) chatbot on Hugging Face LLMs and deployed it on the live site to answer student queries.

International IEEE Conference ICEMCSI — Host — June 2026

PATENT, ACHIEVEMENTS & CERTIFICATIONS

- **Provisional patent filed:** Smart Blind Stick, an AI + IoT assistive cane with obstacle detection and user and caregiver mobile apps.
- **1st place:** E=mc² STEM Expo 2026; UDAAN Intra-Institutional Business Plan Competition 2026; Algo Relay (NHCE MCA Dept.).
- **2nd place:** QuantumX Intercollege Hardware Tech Expo 2026. **3rd place:** Nexgen Convergence Hardware Hustle Tech Expo.
- **Certifications:** Data Analytics (IIT Madras) · AI in Physics (IIT Madras) · C Programming (IIT Bombay)